

IEEE VR 2023 Workshops

Workshop: Mixing Realities: Cross-reality Visualization, Interaction, and Collaboration

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ABSTRACT:

Cross-reality (CR) systems offer different levels of virtuality/physicality to users and enable them to move back and forth between the reality-virtuality continuums in a seamless way. Immersive augmented, virtual, and mixed reality (AR/VR/MR) head-mounted displays (HMDs) have become the main tools that enable cross-reality interaction. However, working with high-dimensional data is challenging due to complex data structures and

their dimensionality. The goal of the workshop is to provide an opportunity for researchers from immersive technologies, HCI, and Visualization fields to share their original ideas and work-in-progress thoughts on the design and evaluation of interactive techniques and systems for effective cross-reality visualization, interaction, and collaboration.

ABOUT THE WORKSHOP:

Immersive analytics has become a rapidly-growing research field with applications in natural sciences in contexts that require users' understanding, exploration, and communication about high-dimensional data. Cross-reality provides users with the possibility of switching visual representations between systems using different degrees of virtuality and allows users to interact with data across multiple technologies (such as AR/VR).

Cross-reality systems have the following four main characteristics: (1) enable multiple realities (virtual $\leftrightarrow$ reality); (2) smooth transition between realities; (3) multiple interaction techniques (touch, gesture); and (4) collaboration between users using systems with multiple realities. However, research on cross-reality systems is still preliminary, and many questions remain to be explored. For instance, to design a cross-reality system, we may need to know users' tasks,

requirements, and mental models (what is expected to be shown in different realities, in which form, and how to enable efficient interaction with the content in each reality).

Through this workshop, we aim to provide a comprehensive overview of the related research on the design and evaluation of interactive techniques and systems for effective cross-reality visualization, interaction, and collaboration. We are interested in discussing technologies merging the virtual and physical worlds to support users in understanding data in different forms. The focus of the workshop is on virtual and augmented reality, and it combines and extends both of these technologies to fully leverage the benefits provided by the four key characteristics of cross-reality approaches in data exploration and content interaction.

ENABLE MULTIPLE REALITIES (VIRTUAL $\leftrightarrow$ REALITY)

Cross-reality merges physical and virtual worlds to create an integrated and seamless user experience [1, 2]. For example, in data visualization, cross-reality enables users to interact with 3D representations of data in a virtual space, where they can manipulate physical objects and receive real-time feedback from their exploration. This enhances their understanding through multiple realities and can improve their engagement and motivation. However, it is crucial to assess the suitability of *data features and their representations* in both physical and virtual environments to communicate information effectively. Additionally, users need the

ability to access information from both realities, compare, and combine them to uncover new insights. Moreover, with new technologies and systems come new opportunities but with them also uncertainties and unknowns. One unknown is about what *tasks and requirements* users have for cross-reality interaction [1, 2]. One focus of the workshop is on discussing topics related to design considerations (data, tasks, and requirements) for cross-reality visualization and representation techniques together with design principles for virtual and physical environments in cross-reality systems.

SMOOTH TRANSITION BETWEEN REALITIES

The *seamless transition between physical and virtual realities* in cross-reality is a vital component for creating an integrated and meaningful user experience [1, 3]. To achieve a smooth transition, several key factors must be considered. Firstly, the design of data representation is crucial in understanding the relationships between objects in both realities. The data representation and user interface should be intuitive, easy to use, and provide clear feedback and visual cues to indicate when a user is transitioning between realities. Secondly, in certain scenarios and tasks, virtual objects are anchored to the physical space, and both 2D and 3D visual representations are

used. *Spatial mapping* is essential in creating a seamless transition between realities. When users move in the physical environment or observe data in the 2D/3D transition region, virtual objects should also move and adjust themselves accordingly. Besides, distortion can affect users' understanding of the data when observing their representation in the 2D/3D region. To explore these topics further, the workshop covers discussions related to cross-reality transitions across multiple technologies, multimodal and cross-modal interaction, perception, and cognition.

MULTIPLE INTERACTION TECHNIQUES (TOUCH, EYE, GESTURE)

Multiple interaction methods are employed in cross-reality systems to enable users to interact with visual and physical representations using gestures, voice commands, eye tracking, touch-based interaction, and tangible interfaces. These technologies offer unique advantages in various scenarios and tasks. For instance, hand and eye tracking involves the use of sensors to monitor a user's hand and eye movements in 3D space, enabling the selection and manipulation of virtual objects by looking at them and reaching out to them [4, 5]. Voice commands allow users to interact with virtual objects by speaking commands and queries. The integration of multiple interaction methods can result in more intuitive, natural, and efficient experiences that mimic our physical world interactions. For instance, integrating gaze and head movement information together could enhance users' selection of objects in AR HMDs, especially when these are small [6]. Similarly, it is possible to leverage body movements that may be underutilized and integrate them to support interactions with other frequently used gestures. [7] shows an

example of using head gestures for mode-switching in VR environments that supplement hand gestures that are already used for the main task of drawing. Such multi-modal interaction techniques have great potential, especially in cross-reality environments that are likely to be complex, and can greatly enhance the user experience and improve the effectiveness of users' interactions.

Cross-reality interaction is a relatively new field, and the technology behind it continues to evolve, particularly with the development of the metaverse. Despite some proposed reviews and frameworks for the metaverse, there is limited research focused specifically on cross-reality interaction, which requires a focus on the technical aspects of immersive devices, networking, tracking systems, input devices, etc. The workshop investigates novel interaction techniques for exploring data in cross-reality environments and covers discussions related to cross-reality interaction and user experience.

COLLABORATION BETWEEN USERS USING SYSTEMS WITH MULTIPLE REALITIES

Research in immersive analytics [8] has highlighted many benefits for data exploration, such as offering spatial perception, creating a profound and impactful representation of data, and facilitating sense-making and collaboration [9, 10, 11]. Cross-reality technologies bring additional potentials for users to interact with the same or different realities simultaneously. Recent research shows that different types of display platforms bring different affordances and engagement when exploring visualizations [12]. When multiple senses, realities, and display platforms are involved, users can feel more deeply connected to their tasks and collaborations. However, there are still many questions that need to be addressed. For example, there are various methods for communication and information sharing, including co-located and synchronous sharing through voice

messages, visual highlights, and shared viewpoints [11, 13]. The complexity of shared information increases when it appears in different realities. The visual representations may be vastly different and difficult for other collaborators, particularly those from different domains, to make sense of. To maximize collaboration, it is crucial to explore an optimized approach that enables individuals from diverse disciplines to focus on their data analysis and exchange insights and findings with effective visualization and interaction techniques. Therefore, in the workshop, we will discuss topics related to collaborative cross-reality immersive analytics and cross-reality applications and productivity.

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